

AI-driven rail capacity planning to manage temporary capacity restrictions under the amended Rail Capacity Regulation annex.

Transportation and Logistics × Production planning and scheduling × Machine learning · OS059 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
59 is the world moving	52 can we play, can we win	MEDIUM 1 named in the evidence	€196m observed evidence · per year	NOW budgeted procurement within 1...	L1 5 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity is an AI-driven rail capacity planning service that helps rail operators and infrastructure managers handle temporary capacity restrictions under the amended EU Rail Capacity Regulation annex. It exists now because the regulation change is imminent and the technology to predict and optimize capacity is mature. The service uses machine learning to model and mitigate the impact of restrictions, reducing disruption and cost.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€314m €97.4m – €1.5bn	€196m €60.7m – €938m	€2.3m €728k – €11.3m	observed
Observed: tendered contract value, annualised	€115m	€115m	€1.4m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Transportation and Logistics (EU27_2020)	113,199 enterprises	observed	Eurostat · sbs_sc_oww	2024
Enterprises adopting Machine learning	2.7 % of enterprises (10+ employees)	observed	Eurostat · isoc_eb_ain2 · E_AI_TML	2025
Annual value of one engagement	€792k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-26..2026-08-05
Size-weighted buyer base	14,748 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

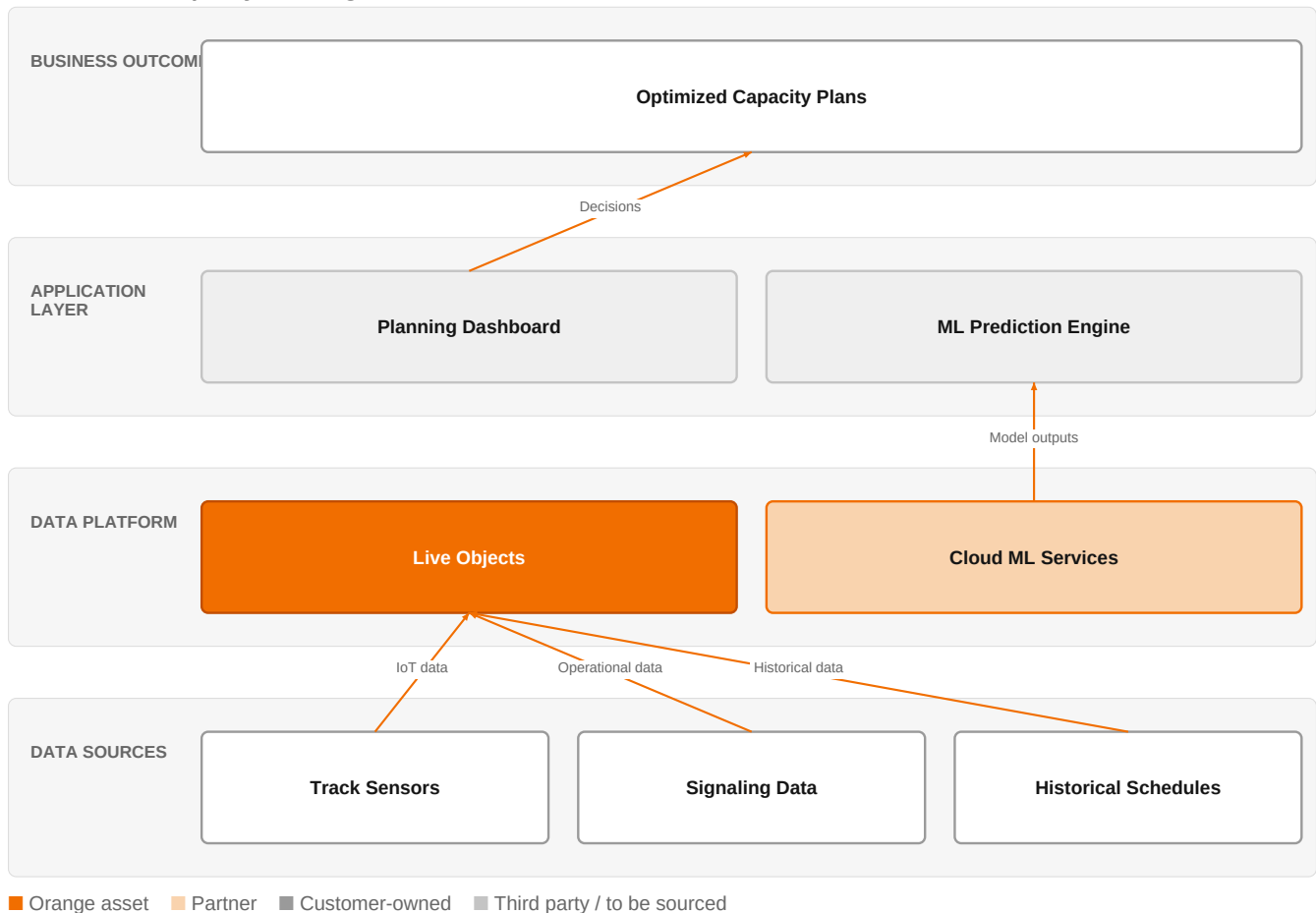
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€115m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-26..2026-08-05
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 8 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

AI-Driven Rail Capacity Planning



This solution turns raw operational data into actionable capacity plans, helping rail operators comply with new regulations and reduce disruption. Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

The EU is amending the Rail Capacity Regulation annex on temporary capacity restrictions, which will force rail operators and infrastructure managers to more rigorously plan and communicate capacity reductions. This regulatory change is the primary driver. Additionally, the technology for AI-driven capacity planning has matured, as evidenced by Microsoft's 'AI Railroad Brain' concept for freight rail, which demonstrates the feasibility of AI in rail operations. The buying behavior is shifting toward predictive, data-driven planning tools.

Sources: SIG-96A1E02831FF ec.europa.eu 2026-05-19; SIG-C765B2D09832 Microsoft 2026-07-16

Who buys, and why now

The buyer is typically the head of network operations or the chief operating officer at a rail infrastructure manager or a large freight operator. The pain is felt by the planning team, who struggle to manually adjust schedules and communicate restrictions across multiple stakeholders. The trigger is the upcoming regulatory deadline, which forces them to demonstrate compliant capacity management. They need to reduce the operational disruption caused by temporary restrictions, such as maintenance windows or incidents, and avoid penalties for non-compliance.

Sources: SIG-96A1E02831FF ec.europa.eu 2026-05-19

Why it is hot now — every claim bound to a dated source

- The EU is amending the Rail Capacity Regulation annex on temporary capacity restrictions, affecting rail operators and infrastructure managers. [SIG-96A1E02831FF]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using its AI/Data/Cloud experts and IoT experts to build a machine learning model that predicts the impact of temporary capacity restrictions on rail traffic. The model would ingest data from the customer's existing systems (e.g., track sensors, signaling) and use Live Objects as the IoT data platform to collect and normalize data. The solution would be deployed on AWS or Google Cloud, leveraging their ML services. The engagement would start with a discovery workshop, then data integration, model development, and finally a dashboard for planners to simulate and optimize capacity allocation.

Why Orange can win

Orange can win because it combines deep expertise in AI/data/cloud with IoT capabilities and a strong partner ecosystem (AWS, Google Cloud, NVIDIA). The reference from Point One Navigation demonstrates Orange's ability to deliver complex location-based solutions. However, the assets are not unique; hyperscalers like Microsoft and AWS also offer ML platforms. Orange's differentiator is its ability to integrate across the customer's entire operational stack, leveraging its network and IoT expertise, and its position as a trusted partner for regulated industries.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Objects	offer	L1	offer provides the technology in a matching domain	unconfirmed
AWS	partner	L2	partner provides the required technology	unconfirmed
Google Cloud	partner	L2	partner provides the required technology	unconfirmed
NiCE	partner	L2	partner provides the required technology	unconfirmed
NVIDIA	partner	L2	partner provides the required technology	unconfirmed
AI / Data / Cloud experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Point One Navigation	reference	SUP	published reference in this vertical, different use case	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 7.1; 1 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitors include Microsoft, which has a strong AI railroad brain concept and is a hyperscaler; AWS and Google Cloud offer ML platforms and are also partners; Atos/Eviden, NiCE, Databricks, Dataiku, and Palantir all sell ML solutions. The customer will compare on the basis of ML sophistication, ease of integration, and industry expertise. Microsoft is strong in AI research and has a rail-specific vision; AWS and Google Cloud are strong in scalable ML infrastructure; Palantir is strong in data integration and decision-making platforms.

Sources: SIG-C765B2D09832 Microsoft 2026-07-16

Competitor	Category	Position here	Basis	Evidence
Microsoft	Hyperscaler	named in this topic's evidence — also an Orange partner	evidenced	SIG-C765B2D09832
AWS	Hyperscaler	sells Machine learning; competes in OX: Smart Industries — also an Orange partner	structural	—
Google Cloud	Hyperscaler	sells Machine learning — also an Orange partner	structural	—
Atos / Eviden	Global systems integrator	sells Machine learning	structural	—
NiCE	Customer-experience platform	sells Machine learning — also an Orange partner	structural	—
Databricks	Data and AI platform	sells Machine learning	structural	—
Dataiku	Data and AI platform	sells Machine learning	structural	—
Palantir	Data and AI platform	sells Machine learning; competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar.
structural means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 How do you currently plan for temporary capacity restrictions, and what is the biggest pain point in that process?
- 2 Have you started preparing for the amended Rail Capacity Regulation annex? What specific compliance requirements are you struggling with?
- 3 What data sources do you have available (e.g., track sensors, signaling data, historical schedules) and how accessible are they?
- 4 Who in your organization would be the primary user of a capacity planning tool, and what decisions would they make with it?
- 5 Have you evaluated any existing AI or ML solutions for capacity planning? What did you find lacking?
- 6 What is your timeline for implementing a new capacity planning process, and is there budget allocated for this year?

Objections you will meet

They will say	You can answer
We already have a team of data scientists who can build this in-house.	That's great, and you may have strong internal capabilities. However, our solution brings pre-built IoT integration and industry-specific expertise that can accelerate deployment and reduce risk. We can also complement your team with our AI/Data/Cloud experts, and you retain full control.
We are already working with Microsoft/AWS/Google on other projects.	That's fine. We partner with all major hyperscalers, so we can build on your existing cloud investments. Our value is in the integration layer and the operational expertise we bring, not in competing with your cloud provider.
The regulation might not affect us directly.	That's a valid concern. The amendment applies to rail operators and infrastructure managers across the EU, so it likely does affect you. Even if the direct impact is small, the ability to predict and manage capacity restrictions can improve operational efficiency and customer satisfaction.

Risks and unknowns

The main risk is that the regulatory amendment may be delayed or changed, reducing urgency. Also, the customer may already have internal analytics teams or existing contracts with hyperscalers. The effectiveness of ML models depends on data quality and availability, which is unknown until a pilot. There is also a risk that the customer perceives Orange as less specialized in rail than competitors like Microsoft or Palantir.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a rail capacity planning solution that leverages Live Objects for IoT data ingestion, AWS or Google Cloud for scalable ML, and NVIDIA for accelerated computing, and include Point One Navigation as a reference for precise positioning in rail contexts.
Sales	In a conversation with a European rail infrastructure manager's head of planning, say: 'We have AI and IoT expertise, and we work with partners like AWS and NVIDIA, to help you model temporary capacity restrictions more accurately under the new EU regulation—could we explore a pilot with your team?'
Strategist / Innovator	Commission a deep-dive brief on how the amended Rail Capacity Regulation annex on temporary capacity restrictions changes planning obligations for rail operators and infrastructure managers, and map where our AI/Data/Cloud and IoT experts plus Orange Group researchers could prototype a capacity-planning solution.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-A3E4FB556CE4	2026-08-17	contractsfinder.servi ce.g...	1	Tunnel Segment Production Monitoring System copy
SIG-8C805BBDC3CA	2026-08-12	contractsfinder.servi ce.g...	1	Tunnel Segment Production Monitoring System
SIG-96A1E02831FF	2026-05-19	ec.europa.eu	1	Amendment of Rail Capacity Regulation annex on temporary capacity restrictions
SIG-2A1D9D3D9ED4	2026-08-10	moneycontrol.com	2	Indian Railways PRS explained : Booking capacity , IRCTC upgrade and key features
SIG-C765B2D09832	2026-07-16	Microsoft	2	The AI Railroad Brain: A new operating model for freight rail - Microsoft

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:14+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:17+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:43:59+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.